

Task 1 - Requirements Analysis

Purpose of the page

The page allows new users to register in Parabank

It collects personal and login information to grant access to account view, fund transfers, and loan requests

Required fields

Field	Type	Required	Description
First Name	Text	Yes	Letters only
Last Name	Text	Yes	Letters only
Address	Text	Yes	May include numbers and spaces
City	Text	Yes	Letters only
State	Text	Yes	2-letter US code
Zip Code	Numeric	Yes	5 digits only
Phone #	Numeric	No	Optional
SSN	Numeric	Yes	9 digits
Username	Text	Yes	Must be unique
Password	Text	Yes	>= 6 characters
Confirm Password	Text	Yes	Must match Password

Validation rules

- 1) All required fields must be completed
- 2) Password and confirmation must match
- 3) Zip Code accepts digits only
- 4) Username must be unique
- 5) Errors appear beside fields in red

- 6) No special characters < > # %
- 7) Password is masked by asterisks

Risks and edge cases

- 1) The existing username error is unclear
- 2) Special chars allowed -> SQL risk
- 3) Weak passwords accepted
- 4) Double registration is possible on lag
- 5) Autofill may break validation
- 6) SSN length not validated
- 7) Password visible if mask fails

Notes

- 1) After registration, a message appears: "Your account was created successfully. You are now logged in."
- 2) User is auto-logged in
- 3) The Account Services menu shows the main options
- 4) Links functional, UI intact
- 5) Password hidden on screen
- 6) Page loads normally without delays

Example valid data

Field	Example
First Name	Dmytro
Last Name	Isai
Address	123 Main Street
City	Los Angeles
State	CA
Zip Code	90001
Phone #	1234567890

SSN	123456789
Username	dmytro_test01
Password	QaTest123!
Confirm	QaTest123!

Task 2 - Test Plan

Objective

To verify that the Parabank registration page functions correctly, users can successfully create new accounts with valid data, and validation handles invalid inputs properly

Environment

Component	Description
Browser	Google Chrome v.129
OS	Windows 11
Device	Laptop (2560×1440)
URL	https://parabank.parasoft.com/parabank/register.htm
Network	Stable Wi-Fi connection

Types of Testing

- 1) Functional Testing - verify that all mandatory fields and the Register button work properly
- 2) UI Testing - ensure all visual elements display correctly
- 3) Validation Testing - client/server validation checks
- 4) Negative Testing - invalid data handling
- 5) Regression Testing - confirm registration still works after fixes

Tools

- 1) Browser DevTools (Chrome)
- 2) Postman (for API checks if applicable)
- 3) Google Sheets for test case management
- 4) Jira / TestCaseLab for reporting

Main Risks

- 1) Missing validation for invalid inputs
- 2) Ability to register a duplicate username
- 3) Data loss on page refresh

- 4) Accessibility issues for screen readers
- 5) Potential SQL injection risks

Notes

The form automatically logs in the user after successful registration, so login verification should also be included in testing

Task 3 - Test Cases

At TestCaseLab, all tests

Notes

Use valid US-style input data (CA, 90001, 123456789)

Cross-browser validation (Chrome, Firefox) and error display behavior should be verified

Task 4 - Bug Report

<https://jira.ithillel.com/secure/RapidBoard.jspa?rapidView=2217&selectedIssue=QM110725-189>

Task 5 - Checklist

Nº	Check Item	Expected Result	Status
1	The login page opens successfully	Page loads without errors	Not Started ▾
2	The “Customer Login” section is visible	Username, Password fields, and the Login button are displayed	Not Started ▾
3	Username field accepts only valid characters (letters/numbers)	No special symbols or spaces accepted	Not Started ▾
4	Password field hides characters with asterisks	Entered text is replaced with dots or stars	Not Started ▾
5	The login button is clickable	Button reacts to click (hover or pressed state)	Not Started ▾
6	“Forgot login info?” link works	Redirects to password recovery page	Not Started ▾
7	“Register” link works	Redirects to registration form	Not Started ▾
8	Empty Username and Password → show error message	“Please enter username and password” appears	Not Started ▾
9	Invalid credentials show an error message	“The username and password could not be verified” appears	Not Started ▾
10	Valid credentials → successful login	User is redirected to Account Overview	Not Started ▾
11	The layout and logo are properly aligned	UI looks consistent with no overlapping	Not Started ▾
12	“Log In” button is disabled after a click until a response is received	Prevents multiple submissions	Not Started ▾

Task 6 - API Testing

file

Task 7 - SQL Queries

1. Customers from 'Los Angeles'

```
SELECT id, first_name, last_name, city
FROM customers
WHERE city = 'Los Angeles';
```

2. Account count per customer

```
SELECT c.id, c.first_name, c.last_name, COUNT(a.id) AS account_count
FROM customers c
JOIN accounts a ON c.id = a.customer_id
GROUP BY c.id, c.first_name, c.last_name;
```

3. Total transactions last month

```
SELECT COUNT(id) AS total_transactions_last_month
FROM transactions
WHERE date >= DATE_TRUNC('month', CURRENT_DATE - INTERVAL '1 month')
      AND date < DATE_TRUNC('month', CURRENT_DATE);
```

Task 8 - Exploratory Testing

#	What was tested	Test Data	Expected Result	Actual Result	Found issues / UX problems	Severity	Priority
1	Loan request with valid numeric data	Loan = 5000 Down = 500 Account = 13566	Loan should be processed — message "Loan approved" or "Denied"	Error! An internal error has occurred and has been logged	Server error even for valid data	High	High
2	Empty fields submitted	Loan = (nyctro) Down = (nyctro) Account = 13566	Should show validation message "Fields are required"	Internal error instead of validation	No front-end validation	Medium	High
3	Non-numeric input	Loan = "abc" Down = (nyctro) Account = 13566	Should show "Invalid input format"	Internal error	Missing input type validation	Medium	Medium
4	Down Payment > Loan Amount	Loan = 1000 Down = 5000 Account = 13566	Should display a logical message "Down payment cannot exceed loan amount"	Internal error	Missing logical validation	Medium	High